

Chapter 9

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Problem 9.1. *Prove the following functions are actually metrics:*

1. $\mathbb{R}$ - $d(x, y) = |x - y|$
2. $\mathbb{R}^n$ - $d(x, y) = \sqrt{(y_1 - x_1)^2 + \cdots + (y_n - x_n)^2}$
3. $\mathbb{R}^2$ - $d(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\}$
4. $\mathbb{R}^2$ - $d(x, y) = |x_1 - y_1| + |x_2 - y_2|$
5. X - $d(x, y) = 1 - \delta(x, y)$
6. L^1, L^2, L^∞ norms on $C[0, 1]$

Solution. 1. Usual norm on $\mathbb{R}$

- (a) $\forall x, y \in \mathbb{R} \ni d(x, y) = 0 \implies d(x, y) = |x - y| = 0 \implies x - y = 0 \implies x = y. \forall x \in \mathbb{R},$
 $d(x, x) = |x - x| = 0$
- (b) $\forall x, y \in \mathbb{R} \ni d(x, y) = |x - y| \geq 0.$
- (c) $\forall x, y \in \mathbb{R}, d(x, y) = |x - y| = |y - x| = d(y, x)$
- (d) $\forall x, y, z \in \mathbb{R},$

$$\begin{aligned} d(x, z) &= |x - z| \\ &= |x - y + y - z| \\ &\leq |x - y| + |y - z| = d(x, y) + d(y, z) \end{aligned}$$

2. Euclidean norm on $\mathbb{R}^n$

- (a) $\forall x, y \in \mathbb{R}^n \ni d(x, y) = 0 \implies \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2} = 0 \implies (x_1 - y_1)^2 + \cdots + (x_n - y_n)^2 = 0 \implies x_i - y_i = 0$ since $(x_i - y_i)^2 \geq 0$. Thus $x_i = y_i \implies x = y. \forall x \in \mathbb{R}^n,$
 $d(x, x) = \sqrt{\sum_{i=1}^n (x_i - x_i)^2} = 0$
- (b) Squares of real values are nonnegative. Sum of positive numbers is nonnegative. Square root of positive number is nonnegative.
- (c)

$$d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2} = \sqrt{\sum_{i=1}^n (y_i - x_i)^2} = d(y, x)$$

(d)

$$\begin{aligned}
d(x, z)^2 &= \sum_{i=1}^n (x_i - z_i)^2 \\
&= \sum_{i=1}^n (x_i - y_i + y_i - z_i)^2 \\
&= \sum_{i=1}^n [(x_i - y_i)^2 + 2(x_i - y_i)(y_i - z_i) + (y_i - z_i)^2] \\
&= d(x, y)^2 + d(y, z)^2 + 2 \sum_{i=1}^n (x_i - y_i)(y_i - z_i) \\
&= d(x, y)^2 + d(y, z)^2 + 2\langle x - y, y - z \rangle \\
&\leq d(x, y)^2 + d(y, z)^2 + 2d(x - y, 0)d(y - z, 0) \\
&= d(x, y)^2 + d(y, z)^2 + 2d(x, y)d(y, z) \\
&= (d(x, y) + d(y, z))^2
\end{aligned}$$

3. Square norm on $\mathbb{R}^2$

(a) $\forall x, y \in \mathbb{R}^2 \ni d(x, y) = 0 \implies \max\{|x_1 - y_1|, |x_2 - y_2|\} = 0 \implies x_1 = y_1 \text{ and } x_2 = y_2$
 $\forall x \in \mathbb{R}^2 \implies d(x, x) = \max(0, 0) = 0$

(b) absolute value is nonnegative. max of nonnegative values is nonnegative.

(c)

$$d(x, y) = \max\{|x_1 - y_1|, |x_2 - y_2|\} = \max\{|y_1 - x_1|, |y_2 - x_2|\} = d(y, x)$$

(d)

$$\begin{aligned}
d(x, z) &= \max\{|x_1 - z_1|, |x_2 - z_2|\} \\
&\leq \max\{|x_1 - y_1| + |y_1 - z_1|, |x_2 - y_2| + |y_2 - z_2|\} \\
&\leq \max\{|x_1 - y_1|, |x_2 - y_2|\} + \max\{|y_1 - z_1|, |y_2 - z_2|\} \\
&= d(x, y) + d(y, z)
\end{aligned}$$

4. Taxicab norm on $\mathbb{R}^2$

(a) $\forall x, y \in \mathbb{R}^2 \ni d(x, y) = 0 \implies |x_1 - y_1| + |x_2 - y_2| = 0 \implies x_1 = y_1 \text{ and } x_2 = y_2$. Thus
 $x = y$. $\forall x \in \mathbb{R}^2, d(x, x) = 0 + 0 = 0$

(b) Absolute value is nonnegative. Sum of nonnegative values is nonnegative.

(c) Pretty trivial

(d)

$$\begin{aligned}
d(x, z) &= |x_1 - z_1| + |x_2 - z_2| \\
&\leq |x_1 - y_1| + |x_2 - y_2| + |y_1 - z_1| + |y_2 - z_2| \\
&= d(x, y) + d(y, z)
\end{aligned}$$

5. Discrete norm on X

- (a) Pretty trivial
- (b) Pretty trivial
- (c) pretty trivial
- (d)

$$d(x, y) = 1 - \delta(x, y) \leq 1 - \delta(x, z) + 1 - \delta(z, y) = d(x, z) + d(z, y)$$

6. L^1 - Triangle inequality

7. L^2

$$\begin{aligned}
d_2(f, g)^2 &= \int_0^1 (f(x) - g(x))^2 dx \\
&= \int_0^1 (f(x) - h(x))^2 + 2(f(x) - h(x))(h(x) - g(x)) + (h(x) - g(x))^2 dx \\
&= d_2(f, h) + d_2(h, g) + 2\langle f - h, h - g \rangle_{L^2} \\
&\leq d_2(f, h) + d_2(h, g) + 2d_2(f - h, 0)d_2(h - g, 0) \\
&= d_2(f, h) + d_2(h, g) + 2d_2(f, h)d_2(h, g) \\
&= (d_2(f, h) + d_2(h, g))^2
\end{aligned}$$

8. L^∞ - Triangle inequality

■

Claim 8.2. X and Y metrizable $\implies X \times Y$ metrizable

Proof. Let $d_{X \times Y}(a, b) = \max(d_x(a_1, b_1), d_y(a_2, b_2))$. $d_{X \times Y}$ is a metric (Proof very similar to square norm on $\mathbb{R}^2$). $\forall W \in \mathcal{T}_{X \times Y}$, $W = \bigcup_{\alpha \in \lambda} (U_\alpha \times V_\alpha)$. $\forall x = (x_1, x_2) \in W$, $x \in U_\alpha \times V_\alpha$ for some $\alpha \in \lambda$. By definition of basis, $\exists a \in X, \epsilon_1 > 0 \ni x_1 \in B_{\epsilon_1}^X(a) \subseteq U_\alpha$. WLOG $a = x_1$. Similarly $x_2 \in B_{\epsilon_2}^Y(x_2) \subseteq V_\alpha$. Thus $x \in B_{\epsilon_1}^X(x_1) \times B_{\epsilon_2}^Y(x_2) \subseteq U_\alpha \times V_\alpha$. $\forall y \in B_{\epsilon_1}^X(x_1) \times B_{\epsilon_2}^Y(x_2)$, $d(x, y) \leq \max(\epsilon_1, \epsilon_2)$. Let $\epsilon = \min(\epsilon_1, \epsilon_2)$. Thus $x \in B_\epsilon^{X \times Y}(x) \subseteq B_{\epsilon_1}^X(x_1) \times B_{\epsilon_2}^Y(x_2) \subseteq U_\alpha \times V_\alpha$. Thus $\mathcal{B}_{d_{X \times Y}}$ is a basis. Thus $X \times Y$ is metrizable. $\square$

Claim 8.3. Prove that every metrizable space is Hausdorff.

Proof. $\forall x, y \in X \ni x \neq y \implies d(x, y) > 0$. Let $\epsilon = \frac{d(x, y)}{2}$. Thus $y \notin B_\epsilon(x) \ni x$ and $x \notin B_\epsilon(y) \ni y$. $\forall z \in B_\epsilon(x) \cap B_\epsilon(y) \implies d(x, z) < \epsilon$ and $d(y, z) < \epsilon$. Thus $d(x, y) \leq d(x, z) + d(z, y) < \epsilon + \epsilon < d(x, y) \implies d(x, y) < d(x, y)$, a contradiction. Thus both open balls are disjoint. Thus X is Hausdorff. $\square$

Claim 8.4. Let $X = C[0, 1]$. Prove $F : X \rightarrow \mathbb{R}$ where

$$F(f) = \int_0^1 f(x) dx$$

is continuous.

Proof. $\forall f \in X, \forall \epsilon > 0$. Let $\delta = \epsilon$. $\forall g \in B_X(f, \delta)$.

$$\begin{aligned} |F(f) - F(g)| &= \left| \int_0^1 (f(x) - g(x)) dx \right| \\ &\leq \int_0^1 |f(x) - g(x)| dx < \delta = \epsilon \end{aligned}$$

Thus $f \in B_X(f, \delta) \subseteq F^{-1}(B_{\mathbb{R}}(F(f), \epsilon))$. Thus, $F^{-1}(B_{\mathbb{R}}(F(f), \epsilon))$ is a open set. Thus F is continuous. $\square$

Problem 8.5. (X, d) is a metrix space. Let $\alpha : X \times X \rightarrow \mathbb{R}$ where $\alpha(x, y) = \min(1, d(x, y))$. Let $\beta : X \times X \rightarrow \mathbb{R}$ where

$$\beta(x, y) = \frac{d(x, y)}{1 + d(x, y)}$$

1. Prove α and β are metrics
2. Prove (X, α) and (X, β) are bounded.
3. Prove α and β generate $\mathcal{T}$

Proof. 1. (a) i. $\forall x, y \in X \ni \alpha(x, y) = 0 \implies d(x, y) = 0 \implies x = y$. If $x = y \implies d(x, y) = 0 \implies \alpha(x, y) = 0$
ii. $\alpha(x, y) = \min(1, d(x, y)) \geq 0$ since $d(x, y) \geq 0$.
iii. $\alpha(x, y) = \min(1, d(x, y)) = \min(1, d(y, x)) = \alpha(y, x)$
iv.

$$\begin{aligned} \alpha(x, z) &= \min(1, d(x, z)) \\ &\leq \min(1, d(x, y) + d(x, z)) \\ &\leq d(x, y) + d(x, z) \end{aligned}$$

- (b) i. $\forall x, y \in X \ni \beta(x, y) = 0 \implies d(x, y) = 0 \implies x = y$. If $x = y \implies d(x, y) = 0 \implies \beta(x, y) = 0$
ii. $\beta(x, y) = \frac{d(x, y)}{1 + d(x, y)} \geq 0$ because $d(x, y) \geq 0$.
iii. $\beta(x, y) = \frac{d(x, y)}{1 + d(x, y)} = \frac{d(y, x)}{1 + d(y, x)} = \beta(y, x)$
iv.

$$\begin{aligned} \beta(x, z) &= \frac{d(x, z)}{1 + d(x, z)} \\ &\leq \frac{d(x, y)}{1 + d(x, y) + d(y, z)} + \frac{d(y, z)}{1 + d(x, y) + d(y, z)} \\ &\leq \beta(x, y) + \beta(y, z) \end{aligned}$$

2. (a) $\alpha(x, y) = \min(1, d(x, y)) \leq 1$

- (b) $\beta(x, y) - 1 = \frac{-1}{1+d(x, y)} \leq 0 \implies \beta(x, y) \leq 1$
3. (a) $\forall U \in \mathcal{T}, \forall x \in U, \exists \epsilon > 0 \ni x \in B_d(x, \epsilon) \subseteq U$. Let $\delta = \min(0.5, \epsilon)$. $\forall y \in B_\alpha(x, \delta)$, $\alpha(x, y) < \min(1, d(x, y)) < \delta \leq \epsilon \implies d(x, y) < \epsilon$. Thus $x \in B_\alpha(x, \delta) \subseteq B_d(x, \epsilon) \subseteq U$. Thus α generates a basis for $\mathcal{T}$. Thus α generates $\mathcal{T}$.
- (b) $\forall U \in \mathcal{T}, \forall x \in U, \exists \epsilon > 0 \ni x \in B_d(x, \epsilon) \subseteq U$. Let $\delta = \frac{\epsilon}{1+\epsilon}$. Thus $\forall y \in B_\beta(x, \delta)$, let $d(x, y) = u$

$$\begin{aligned}\beta(x, y) &= \frac{u}{1+u} < \frac{\epsilon}{1+\epsilon} \\ \implies u + \epsilon u &< \epsilon + \epsilon u \\ \implies u &< \epsilon\end{aligned}$$

Thus $x \in B_\beta(x, \delta) \subseteq B_d(x, \epsilon) \subseteq U$. Thus β generates a basis for $\mathcal{T}$. Thus β generates $\mathcal{T}$. $\square$

Claim 9.6. (X, α) and (Y, β) are metric spaces. Let $f : X \rightarrow Y$ be a surjective isometry. Prove f is a homeomorphism

Proof. $\forall x, y \in X \ni f(x) = f(y) \implies \beta(f(x), f(y)) = 0 \implies \alpha(x, y) = 0$, by definition of isometry. Thus $x = y$. Thus f is injective.

$\forall x \in X$ and $\forall \epsilon > 0$, $f(B_X(x, \epsilon)) \subseteq B_Y(f(x), \epsilon)$. $\forall y \in B_Y(f(x), \epsilon)$, by surjectivity and definition of isometry, $\exists a \in B_X(x, \epsilon) \ni f(a) = y$. Thus $f(B_X(x, \epsilon)) = B_Y(f(x), \epsilon)$. Thus f is an open function.

$\forall y \in Y$ and $\forall \epsilon > 0$, by injectivity $\exists! x \in X \ni f(x) = y$ and $f^*(B_Y(y, \epsilon)) \subseteq B_X(x, \epsilon)$, since f is an isometry. Since $f(B_X(x, \epsilon)) = B_Y(y, \epsilon) \implies B_X(x, \epsilon) \subseteq f^*(B_Y(y, \epsilon))$. Thus f is continuous by 7.13.

By 7.28, f is a continuous open bijection thus f is a homeomorphism. $\square$

Claim 7. X is a metric space with metric d . For nonempty subsets A and B and for $x \in X$, define

$$\begin{aligned}d(x, A) &= \inf \{d(x, a) \mid a \in A\} \\ d(A, B) &= \inf \{d(a, b) \mid a \in A, b \in B\}\end{aligned}$$

1. Show both are well defined
2. Show that if $A \cap B \neq \emptyset$ then $d(A, B) = 0$
3. Give an example that the previous implication does not reverse
4. Show that these definitions are connected in the following ways:

- (a) $d(x, A) = d(\{x\}, A)$
- (b) $d(A, B) = \inf \{d(a, B) \mid a \in A\} = \inf \{d(b, A) \mid b \in B\}$

5. Show that $A \subseteq X$ is closed (in the topology generated by d) if and only if $d(x, A) \neq 0$ for all $x \in X - A$.

Proof. 1. $\{d(x, a) \mid a \in A\}$ is bounded below by 0. Thus $d(x, A) = \inf \{d(x, a) \mid a \in A\}$ exists. Same for $d(A, B)$.

2. Suppose $A \cap B \neq \emptyset \implies \exists x \in A \cap B$, thus $0 \in \{d(a, b) \mid a \in A, b \in B\}$. $\forall x, y \in X, d(x, y) \geq 0$. Thus $\forall (a, b) \in A \times B \implies d(a, b) \geq 0$. Thus 0 is a lower bound and is the greatest lower bound. Thus $d(A, B) = 0$.
3. $A = (0, \infty)$ and $B = (-\infty, 0)$ for $\mathbb{R}$
4. $d(x, A) = \inf \{d(x, a) \mid a \in A\} = \inf \{d(b, a) \mid b \in x, a \in A\} = d(\{x\}, A)$.
 $d(A, B) = \inf \{d(a, b) \mid a \in A, b \in B\} = \inf \{\inf \{d(a, b) \mid b \in B\} \mid a \in A\} = \inf \{d(a, B) \mid a \in A\}$
5. Suppose $\exists x \in X - A \ni d(x, A) = 0$. Thus $\forall \epsilon > 0, \exists a \in A \ni d(x, a) < \epsilon$ by definition of infimum. Thus $a \in B_\epsilon(x)$. Thus $B_\epsilon(x) \cap A \neq \emptyset$. Thus $\forall U \ni x \in U, U \cap A \neq \emptyset$ since $B_\epsilon(x) \subseteq U$ for some ϵ . Thus A is not closed. Thus by contrapositive if A is closed then $\forall x \in X - A, d(x, A) \neq 0$. Assume the contrary that A is not closed. Thus $\exists x \in \bar{A} - A$. Thus $d(x, A) = \epsilon > 0$. Let $\delta = \frac{\epsilon}{2}$. Thus $\forall a \in A, d(x, a) \geq \epsilon > \frac{\epsilon}{2} \implies a \notin B_\delta(x)$. Thus $x \notin A$. Which is a contradiction. Thus $\bar{A} = A$. A is closed. □

Claim 9.8. *Let X be a metrizable space. X is regular*

Proof. Let d be the metric which generates $\mathcal{T}$. $\forall A \in \mathcal{C} \ni A \neq X$ and $\forall x \in X - A$. $\exists \epsilon > 0 \ni x \in B_\epsilon(x) \subseteq X - A$. Let $V = \bigcup_{a \in A} B_{\frac{\epsilon}{2}}(a)$ and $U = B_{\frac{\epsilon}{2}}(x) \subseteq B_\epsilon(x)$.

Assume the contrary U and V aren't disjoint. Thus $\exists y \in U \cap V$. Thus $d(x, y) < \frac{\epsilon}{2}$ and $d(y, a) < \frac{\epsilon}{2}$ for some $a \in A$.

$$\begin{aligned} d(x, a) &\leq d(x, y) + d(y, a) \\ &< \frac{\epsilon}{2} + \frac{\epsilon}{2} < \epsilon \end{aligned}$$

Thus $a \in U$ which is a contradiction since $U \subseteq X - A$. Thus $U \cap V = \emptyset$ and X is regular □

Claim 9.10. *Let X be a metrizable space. X is normal*

Proof. Let d be the metric which generates $\mathcal{T}$. $\forall A, B \in \mathcal{C} \ni A \cap B = \emptyset$. $\forall x \in A, \exists \epsilon_x > 0 \ni x \in B_{\epsilon_x}(x) \subseteq X - B$. $\forall y \in B, \exists \delta_y > 0 \ni y \in B_{\delta_y}(y) \subseteq X - A$. Note $\epsilon_x \leq d(x, B) \leq d(x, b)$ and $\delta_y \leq d(y, A) \leq d(y, a)$.

$$\begin{aligned} U &= \bigcup_{x \in A} B_{\frac{\epsilon_x}{2}}(x) \\ V &= \bigcup_{y \in B} B_{\frac{\delta_y}{2}}(y) \end{aligned}$$

Assume the contrary that $U \cap V \neq \emptyset$. $\exists y \in U \cap V \implies \exists a \in A, \exists b \in B \ni d(y, a) < \epsilon_a, d(y, b) < \delta_b$. Thus,

$$\begin{aligned}
d(a, b) &\leq d(a, y) + d(y, b) \\
&< \frac{\epsilon_a}{2} + \frac{\delta_b}{2} \\
&\leq \frac{d(a, b)}{2} + \frac{d(a, b)}{2} \\
&= d(a, b)
\end{aligned}$$

Which is a contradiction. Thus $U \cap V = \emptyset$. Thus X is normal. $\square$

Claim 9.11. *Let $(X, \mathcal{T})$ be a metrizable, topological space with the countable chain condition. Let d be a metric on X that generates the topology $\mathcal{T}$. All ϵ -balls we refer to below will be with respect to this metric.*

1. Fix $n \in \mathbb{N}$. Show that there exists a maximal collection of mutually disjoint $\frac{1}{n}$ -balls in X . (Maximal with respect to inclusion. That is, show that there is a collection B_n of mutually disjoint $\frac{1}{n}$ -balls in X that is not properly contained in any larger such collection.) (Hint: Use Zorn's Lemma.)
2. For each $n \in \mathbb{N}$, suppose B_n is a collection as in the previous part. Let $C_n \subseteq X$ be the collection of centres of the balls in B_n , and finally let $C = \bigcup_{n \in \mathbb{N}} C_n$. Show that C is a countable, dense subset of X . Conclude that X is separable.
3. Now that we know that X is separable. Prove that every separable metrizable space is second countable. (Hint: Use the fact that every point in a metric space has a particularly nice countable local basis.)

Proof. 1. Let P be the set of all collections of $\frac{1}{n}$ -balls of X , P is a poset with a partial ordering of inclusion. $\left\{ B_{\frac{1}{n}}(x) \right\}_{x \in X}$ is the upper bound of any chain in P . Let A be the subset of P , of collections of disjoint $\frac{1}{n}$ balls. By Zorn's Lemma, A has a maximal element. Let B_n be the maximal element of A . Thus B_n is the maximal collection of disjoint $\frac{1}{n}$ -balls.

2. C is a countable union of countable sets, thus C is countable. Assume the contrary C is not dense. Thus $\exists U \in \mathcal{T} - \{\emptyset\} \ni U \cap C = \emptyset$. For $x \in U$, $\exists \epsilon > 0 \ni x \in B_\epsilon(x) \subseteq U \implies B_\epsilon(x) \cap C = \emptyset$. $\exists n \in \mathbb{N} \ni \epsilon > \frac{2}{n} \implies x \in B_{\frac{2}{n}}(x) \subseteq B_\epsilon(x) \subseteq U$. $B_{\frac{2}{n}}(x) \cap C = \emptyset$. Thus for all $c \in C_n$, $B_{\frac{2}{n}}(x) \cap B_{\frac{1}{n}}(c) = \emptyset$. Thus $B_n \cup B_{\frac{1}{n}}(x)$ is a disjoint collection of $\frac{1}{n}$ balls, contradicting the maximality of B_n . Thus by contradiction C is dense.

3. Let $\mathcal{B} = \bigcup_{n \in \mathbb{N}} B_n$. $\forall U \in \mathcal{T}$ and $\forall x \in U$, $\exists n \in \mathbb{N} \ni x \in B_{\frac{2}{n}}(x) \subseteq U$. By definition of density, $\exists y \in B_{\frac{1}{n}}(x) \cap C$. Thus, $\forall z \in B_{\frac{1}{n}}(x)$

$$\begin{aligned}
d(y, z) &\leq d(x, y) + d(x, z) \\
&< \frac{1}{n} + \frac{1}{n} = \frac{2}{n}
\end{aligned}$$

Thus $x \in B_{\frac{1}{n}}(y) \subseteq B_{\frac{2}{n}}(x) \subseteq U$. $\square$

Claim 9.12a. Let X be a set and (Y, d) a metric space. Suppose $\{f_n\}_{n \in \mathbb{N}}$ is a sequence of functions $f_n : X \rightarrow Y$ that converges uniformly to a f . Prove that $\{f_n\}_{n \in \mathbb{N}}$ converges pointwise to f .

Proof. $\forall x \in X$, and $\forall U \in \mathcal{T}_Y \ni f(x) \in U$, $\exists \epsilon > 0 \ni B_\epsilon(f(x)) \subseteq U$. By definition of uniform convergence, $\exists K \in \mathbb{N} \ni \forall n \geq K, f_n(x) \in B_\epsilon(f(x)) \subseteq U$. Thus by definition of limit of sequence $f(x)$ is the limit of $f_n(x)$. Thus f_n converges pointwise to f . $\square$

Problem 9.12b. Give an example of a sequence of functions which converges pointwise to 0 but not uniformly

Proof. Let

$$f_n(x) := \begin{cases} (n+2)x & x \in [0, \frac{1}{n+2}) \\ -(n+2)x - 2 & x \in [\frac{1}{n+2}, \frac{2}{n+2}) \\ 0 & x \in [\frac{2}{n+2}, 1] \end{cases}$$

$f_n \rightarrow 0$ pointwise but not uniformly. $\square$

Claim 9.12c. If $f_n \rightarrow f$ uniformly where f_n is continuous then f is continuous

Proof. $\forall U \in \mathcal{T}_Y, \forall x \in f^{-1}(U)$. $\exists \epsilon > 0 \ni x \in f^*(B_\epsilon(f(x))) \subseteq f^{-1}(U)$.
 $\exists K \in \mathbb{N} \ni \forall n \geq K, |f_n - f|_\infty < \frac{\epsilon}{3}$.
 $\forall x \in X, \forall a \in f_K^*(B_\epsilon(f(x)))$

$$\begin{aligned} d(f(x), f(a)) &\leq d(f(x), f_n(x)) + d(f_n(x), f_n(a)) + d(f_n(a), f(a)) \\ &< \frac{\epsilon}{3} + \frac{\epsilon}{3} + \frac{\epsilon}{3} \\ &= \epsilon \end{aligned}$$

Thus, $x \in f_K^*(B_{\frac{\epsilon}{3}}(f(x))) \subseteq f^*(B_\epsilon(f(x))) \subseteq U$ $\square$

Problem 9.13a. Example of a metric space and function f for which $d(f(x), f(y)) < d(x, y)$ but f is not a contraction mapping

Proof. Let $X = [0, \infty)$ with standard metric. $f(x) = \log(1+x)$ is not a contraction mapping. $\square$

Claim 9.13b. Every contraction mapping has at most 1 fixed point

Proof. Assume the contrary, that f is a contraction mapping more than 1 fixed points. $\exists x, y \in X \ni x \neq y$ and $f(x) = x$ and $f(y) = y$. Thus $d(x, y) = d(f(x), f(y)) < \alpha d(x, y)$ for $\alpha \in (0, 1)$. Thus $d(x, y) < \alpha d(x, y) < d(x, y)$ since $d(x, y) \geq 0$ and $\alpha \in (0, 1)$. This is a contradiction. Thus by contradiction, f has at most 1 fixed point. $\square$

Claim 9.14. Let F be a finite set of points in $\mathbb{R}^n$. Prove there is a unique closed ball of minimal radius which encloses F .

Proof. Note that $B_\epsilon(x)$ will be the closed ball of radius ϵ centered at x .

Let P be the set of all closed balls of $\mathbb{R}^n$ with the partial ordering of size of radius (the smaller the size of the radius, the "greater" the ball). Let A be the set of all balls which enclose F . For any chain in A , the upper bound is a ball whose radius is less than $\min(d(f, g) \mid f, g \in F \ni f \neq g)$, thus the ball doesn't enclose F . Thus by Zorn's lemma A has a maximal element. Thus there exists a ball B which encloses F with minimal radius.

Let $m(x) = \max \{d(f, x) \mid f \in F\}$. $\forall x \in \mathbb{R}^n, \forall f \in F, d(f, x) \leq m(x) \implies f \in B_{m(x)}(x)$. For all $r < m(x)$, by definition of maximum, $\exists f \in F \ni d(f, x) = m > r$. Thus for all points $x \in X$, the ball of radius $m(x)$ is the ball of minimal radius centered at x such that the ball completely encloses F .

Let B be one of the maximal elements of P such that the radius of B is minimal. Let c be the center of B . Thus the radius of B is the minimum value of m for all $\mathbb{R}^n$. Let M , be the radius of B . By the above $M = \min \{m(x) \mid x \in \mathbb{R}^n\} = m(c)$.

Assume the contrary, $\exists d \in \mathbb{R}^n \setminus c \ni m(x) = M$. Thus $F \subseteq B_M(c) \cap B_M(d)$. Thus $d(c, d) \leq 2M$. $u = \frac{c+d}{2} \in B_M(c) \cap B_M(d)$. Let $a = \frac{d(c, d)}{2}$. Let $r = \sqrt{M^2 - r^2}$. Note $r < M$. Thus $\forall x \in B_M(c) \cap B_M(d)$.

$$d(u, x) =$$

□

Claim 9.14. *Let (X, d) be a complete metric space. If $f : X \rightarrow X$ is a contraction. Then f has a fixed point*

Proof. Let $f_n : X \rightarrow X$ where $f_n = f \circ f_{n-1}$ where $f_1 = f$. $\forall x \in X, d(f_2(x), f(x)) \leq \alpha d(f_1(x), x) \implies d(f_{n+1}(x), f_n(x)) \leq \alpha^n d(f(x), x)$. Let $x_n = f_n(x)$ and let $x_0 = x$. WTS x_n is cauchy.

$\forall \epsilon > 0$. By the archimedean property, $\exists K \in \mathbb{N} \ni K < \log_\alpha \left(\frac{\epsilon(1-\alpha)}{d(x_1, x_0)} \right)$. Thus $\forall m, n \geq K$,

$$\begin{aligned} d(x_m, x_n) &\leq d(x_m, x_{m-1}) + \cdots + d(x_{n+1}, x_n) \\ &= \sum_{k=n}^{m-1} d(x_{k+1}, x_k) \\ &\leq \sum_{k=n}^{m-1} \alpha^k d(x_1, x_0) \\ &= \alpha^n d(x_1, x_0) \sum_{k=0}^{m-n-1} \alpha^k \\ &< \alpha^n d(x_1, x_0) \sum_{k=1}^{\infty} \alpha^k \\ &\leq \frac{\alpha^K d(x_1, x_0)}{\alpha - 1} < \epsilon \end{aligned}$$

Thus x_n is cauchy. By completeness of X , $\exists x \ni x_n \rightarrow x^*$. By continuity of f , $f(x^*) = f(\lim_{n \rightarrow \infty} x_n) = \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x^*$. Thus x^* is a fixed point. □

Claim 9.15. *Let X be a complete metric space, and let $\{U_i\}_{i \in \mathbb{N}}$ be a collection of open dense subsets of X . Then $\bigcup_{i \in \mathbb{N}} U_i$ is a dense subset of X*

Proof. $\forall U \in X, \exists x_1 \in U \cap U_1$ by definition of a dense subset.

$\exists 0 < r_1 < 1 \ni x_1 \in \overline{B_{r_1}(x_1)} \subseteq U \cap U_1$. By definition of dense subset, $\exists x_2 \in B_{r_1}(x_1) \cap U_2 \subseteq U \cap U_1 \cap U_2$. $\exists 0 < r_2 < \frac{1}{2} \ni x_2 \in \overline{B_{r_2}(x_2)} \subseteq B_{r_1}(x_1) \cap U_2$. Thus $B_{r_1}(x_1) \cap U_2 \supseteq \overline{B_{r_2}(x_2)}$. This can be done recursively, to find a pair of sequences x_n and $0 < r_n < \frac{1}{n}$ where

$$\overline{B_{r_n}(x_n)} \subseteq B_{r_{n-1}}(x_{n-1}) \cap U_n$$

Thus x_n is Cauchy and $\exists x$ such that $x_n \rightarrow x$ because X is complete. Since X is metrizable thus 1st countable, $x \in \overline{B_{r_n}(x_n)}$ for all $n \in \mathbb{N}$. Since $\overline{B_{r_n}(x_n)} \subseteq U_n \cap U \implies x \in U_n \cap U$ for all n . Thus $x \in U \cap (\bigcup_{n \in \mathbb{N}} U_n)$. Thus $\bigcup_{n \in \mathbb{N}} U_n$ is dense. $\square$